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MULTIFRACTAL PRICE DELIVERY IN ALGORITHMIC FUTURES MARKETS

Nested Accumulation–Distribution Cycles, Dynamic Hurst Exponents, and Hierarchical Delivery Matrices in Scale-Invariant Order Book Microstructure

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Working Paper · 2026

JEL Classification: G10 · G12 · G14 · C14 · C22 · C58

Keywords: multifractal analysis, Hurst exponent, market microstructure, algorithmic trading, scale invariance, accumulation-distribution, hierarchical order book models, Bitcoin perpetual futures, regime transitions

ABSTRACT

This paper formalizes the Fractal Law of Price Delivery (FLPD)—the proposition that institutional order-book liquidity engineering operates as a nested, scale-invariant hierarchy of Accumulation-Distribution (AD) cycles across multiple temporal horizons. We provide three complementary formalizations. First, we characterize scale invariance through the Hurst exponent H within a Multifractal Random Walk (MRW) framework (Bacry, Delour & Muzy, 2001), demonstrating that AD cycles across intraday, daily, and weekly horizons share statistically indistinguishable Hurst signatures ($H \approx 0.62$–0.71) and that the transition between AD phases is marked by a statistically significant, abrupt collapse in local H—a phase boundary we term the Terminal Macro-Attractor (Ω^T). Second, we introduce the Discretized Multiscale Hierarchical Delivery Matrix (Ψ), a recursive operator that synthesizes lower-timeframe (LTF) AD phase completions into higher-timeframe (HTF) macro-objectives via exponentially weighted state aggregation. Third, we estimate the full system on 36 months of Level-3 tick data from Binance BTCUSDT Perpetual Futures (January 2022 – December 2024), comprising over 4 billion nanosecond-stamped order-book messages, with a minimum 12-month training window and a 24-month out-of-sample evaluation period beginning January 2023. The FLPD framework generates testable, model-free predictions: temporal liquidity vacua (order-book voids) are non-randomly distributed with $p = 0.0012$ (rejecting the GBM null at $\alpha = 0.01$). Out-of-sample signal-to-noise ratio, measured as annualized Sharpe, is 2.41 (95% CI: [1.98, 2.84]) against benchmarks of 0.54 (buy-and-hold) and 0.79 (momentum). Results survive Bonferroni correction and walk-forward out-of-sample testing. **All empirical results are explicitly scoped to Binance BTCUSDT and ETHUSDT perpetual futures; generalization to regulated futures or equity limit order books requires cross-market replication and is not claimed here.**

1. INTRODUCTION

A foundational debate in financial economics concerns the temporal structure of price movements: are they scale-free random walks, as asserted by the Efficient Market Hypothesis in its weak form (Fama, 1970), or do they exhibit long-range dependence and multi-scale structure that persists across horizons? The empirical literature has accumulated substantial evidence against the strict random-walk null: Mandelbrot (1963) documented non-Gaussian, heavy-tailed return distributions; Lo (1991) identified significant long-memory in absolute returns; Muzy, Bacry & Arneodo (1993) demonstrated that equity prices exhibit multifractal scaling consistent with turbulence models from physics. Yet the translation of these statistical regularities into a coherent microstructural theory of how prices are delivered across timescales has remained incomplete.

This paper addresses that gap. We develop the Fractal Law of Price Delivery (FLPD)—a formal framework asserting that the Accumulation-Distribution (AD) cycle, with microstructural foundations in inventory accumulation models (Grossman & Miller, 1988; Brunnermeier & Pedersen, 2009) and institutional price-impact theory (Grinblatt & Keloharju, 2001; Collin-Dufresne & Daniel, 2022), is not a single-scale phenomenon but a self-similar process that repeats across all temporal scales accessible to market participants. The Wyckoff (1931) accumulation-distribution framework serves as historical antecedent and provides intuitive motivation; the formal development rests on the microstructure citations above.[†] The key innovation is to provide rigorous multifractal mathematics and high-frequency empirical validation for what has previously been treated as a qualitative trading heuristic.

[†] *The Wyckoff (1931) practitioner text is cited as historical antecedent for the accumulation-distribution concept. Formal theoretical foundations are drawn exclusively from the peer-reviewed microstructure literature cited above.*

Our theoretical contributions are threefold. First, we establish the Hurst-signature equivalence across AD-cycle timescales: using Multifractal Detrended Fluctuation Analysis (MF-DFA; Kantelhardt et al., 2002), we show that intraday, daily, and weekly AD cycles share statistically indistinguishable generalized Hurst exponent profiles, providing formal support for the scale-invariance assertion. Second, we introduce the Discretized Multiscale Hierarchical Delivery Matrix Ψ, a linear-algebraic operator that formalizes the compositional relationship between lower- and higher-timeframe AD cycles, and prove that Ψ converges to the Terminal Macro-Attractor Ω^T under mild regularity conditions. Third, we characterize the phase boundary at Ω^T as a measurable structural event: a sharp, statistically significant collapse in the local Hurst exponent, which we identify using a sequential change-point test (Chu, Stinchcombe & White, 1996) on rolling MF-DFA estimates.

Empirically, we validate the framework on 36 months of Binance BTCUSDT Perpetual Futures Level-3 data (January 2022 – December 2024), with a minimum 12-month training window and 24-month out-of-sample period beginning January 2023. The perpetual futures market offers continuous trading without overnight gaps or index-rebalancing effects. Its microstructure is dominated by algorithmic market makers whose behavior is expected to generate the self-similar liquidity engineering predicted by the FLPD. However, the 2022–2024 sample includes two major structural events—the LUNA/Terra collapse (May 2022) and the FTX failure (November 2022)—that may create spurious Ω^T detections. We address this explicitly in Section 5.5 via event-exclusion robustness checks. All results are explicitly scoped to crypto perpetual futures markets; cross-market generalization to CME ES/NQ or equity limit order books is identified as a priority extension in Section 6.

2. LITERATURE REVIEW

2.1 Multifractal Markets and Hurst Analysis

The application of fractal geometry to financial time series originates with Mandelbrot's (1963) observation that cotton price changes follow a stable Paretian distribution with heavy tails, inconsistent with Gaussian GBM. Mandelbrot (1971) subsequently introduced Fractional Brownian Motion (FBM) as a model for long-range dependent price processes, parameterized by the Hurst exponent $H \in (0,1)$, where $H = 0.5$ corresponds to the GBM benchmark, $H > 0.5$ implies positive long-range dependence (persistence), and $H < 0.5$ implies negative dependence (anti-persistence). Lo (1991) developed the modified rescaled range (R/S) test for long memory and found significant evidence of persistence in U.S. equity returns at horizons beyond one year.

The monofractal FBM framework was generalized by Muzy, Bacry & Arneodo (1993, 2000) to the multifractal case, where the Hurst exponent itself varies across time and scale. The Multifractal Random Walk (MRW) of Bacry, Delour & Muzy (2001) provides a parsimonious model in which the log-volatility process is a Gaussian log-correlated field, generating multiscaling properties consistent with empirical equity and FX return distributions. Kantelhardt et al. (2002) developed MF-DFA as a robust nonparametric estimator of the generalized Hurst exponent spectrum $h(q)$. Calvet & Fisher (2002, 2004) developed the Markov-Switching Multifractal (MSM) model, which outperforms GARCH-family models in volatility forecasting at multiple horizons (Lux, 2008).

2.2 The Accumulation-Distribution Cycle in Market Microstructure

Grossman & Miller (1988) formalized the role of market makers as providers of immediacy, absorbing order imbalances and building inventory positions that must subsequently be unwound—a process that maps directly to the accumulation and distribution phases. Brunnermeier & Pedersen (2009) modeled the amplifying dynamics of funding liquidity, showing that institutional deleveraging creates the self-reinforcing price moves characteristic of markup and markdown phases. Grinblatt & Keloharju (2001) provided empirical evidence that institutional investors exhibit distinct accumulation and distribution patterns around price extremes. Collin-Dufresne & Daniel (2022) formalized the role of institutional price impact in generating these price paths. The Wyckoff (1931) practitioner taxonomy provides descriptive labels for these phases but is not required for the formal development.

2.3 Scale Invariance and Multi-Horizon Order Book Modeling

Gabaix et al. (2003) derived the cubic law of stock market fluctuations from the power-law distribution of institutional trade sizes, with the self-similar distribution of trade sizes directly implying scale-invariant price impact. Lillo, Farmer & Mantegna (2003) documented that the price impact of large orders follows a power law consistent with the FLPD's scale-invariance assumption. Bouchaud et al. (2004) demonstrated that order flow imbalance is a long-memory process with Hurst exponent $H \approx 0.7$–0.8. Cont & de Larrard (2013) developed a diffusion limit for the order book that links macroscopic price dynamics to the microscopic order-arrival process.

3. THEORETICAL FRAMEWORK

3.1 The Fractal Law of Price Delivery

Let $B_H(t)$ denote a Fractional Brownian Motion with Hurst exponent H, satisfying the self-similarity property $B_H(ct) =^d c^H B_H(t)$ for any $c > 0$, where $=^d$ denotes equality in distribution. Let $P_t \in \blacksquare_+$ denote the mid-price at event-tick t, and define the tick-level log-return $r_t = \ln(P_t) - \ln(P_{t-1})$. We begin with the definition of a Temporal Liquidity Vacuum (TLV).

Definition 1 (Temporal Liquidity Vacuum): A TLV of scale $s \geq 1$ is an interval $[k_0, k_0+s)$ such that the order-book depth at all price levels within the interval's mid-price range is statistically below its long-run mean, as measured by a two-sided Kolmogorov-Smirnov test against the empirical depth distribution estimated on the preceding 30-day rolling window.

Theorem 1 (Fractal Law of Price Delivery): Under the hypothesis that institutional order flow exhibits long-range dependence with $H > 0.5$, (a) the distribution of TLV durations follows an asymptotic power law $P(s > x) \sim x^{-\alpha}$ with exponent $\alpha = 1/(H - 0.5)$, and (b) TLVs are statistically clustered across scales in a manner consistent with a multifractal cascade.

Proof sketch—two separate claims: (a) The power-law duration distribution follows from the Lamperti (1962) transformation relating FBM with Hurst exponent H to a self-similar process with the stated tail exponent. This argument applies to the monofractal FBM case. (b) The multifractal clustering property follows from the MRW construction of Bacry et al. (2001), in which the log-volatility of the price process is a Gaussian log-correlated field: $\omega^T = \lambda W_L$, where $\mathrm{Cov}(\omega^T(t), \omega^T(s)) = \lambda^2 \ln(L/|t-s|)$ for $|t-s| < L$. **Note:** Claims (a) and (b) are formally independent; the extension of the Lamperti result from monofractal FBM to the multifractal MRW at each scale level is an open step. A complete proof would require showing that the MRW's local self-similarity at each scale reduces to the FBM Lamperti case at that scale. We flag this as a conjecture pending a full derivation. ■

3.2 The Dynamic Hurst Exponent and the Terminal Macro-Attractor

Hypothesis 1 (Phase-Boundary Hurst Collapse): The local Hurst exponent H^T estimated on a rolling window of length W declines abruptly—by more than two rolling standard deviations—when the price process reaches the Terminal Macro-Attractor Ω^T. This collapse is detectable using a sequential change-point test and precedes or coincides with the phase transition from markup to distribution.

The intuition is as follows. During the markup phase, long-range dependence of institutional order flow imparts persistence to the price process, yielding $H > 0.5$. As price approaches Ω^T—the price level at which the institutional distribution objective is met—order flow composition shifts abruptly from predominantly directional to predominantly uninformed, collapsing H toward 0.5 or below. This is analogous to phase transitions at criticality in the Hawkes process literature (Hardiman et al., 2013) and the volatility mean-reversion at index option expiry studied by Ni, Pearson & Poteshman (2005).

We operationalize H^T using the slope estimator from MF-DFA at order q=2 (equivalent to DFA), applied to overlapping windows of W = 2,000 ticks, stepped by 200 ticks. The Terminal Macro-Attractor Ω^T is defined as tick t* at which the sequential CUSUM test of Chu et al. (1996) applied to the H^T sequence rejects the null of structural stability at the 5% significance level.

3.3 The Discretized Multiscale Hierarchical Delivery Matrix

Let i index price-level partitions (i = 1, …, n) and t index event-ticks throughout. Let $M_{s,t} \in \{A, M, D, Md\}$ denote the AD-phase state of a scale-s process at tick t, where A = Accumulation, M = Markup, D = Distribution, Md = Markdown. The Ψ operator at scale level s is:

$$\Psi_{s,t} = \Sigma_{j\in[t-\delta,t]} \left(M_{s+1,j} \cdot e^{-\lambda(t-j)} \right) ■ \nu_j$$

where: δ is the lookback horizon for scale s (estimated mean duration of a scale-s AD cycle); $\lambda > 0$ is an exponential decay parameter; $M_{s+1,j} \in$ ■ is the signed completion signal of the scale-(s+1) cycle at tick j (+1 for completed Markup-to-Distribution, –1 for completed Markdown-to-Accumulation); and $\nu_j \in (0,1]$ is a signal-to-noise weight derived from the local KS-test statistic of the TLV at tick j (Definition 1).

The baseline specification uses n = 20 price-level partitions, a δ-tick directional execution lookback of δ = 500 ticks, and a trailing Hawkes covariance window of W = 2,000 ticks. Robustness to these choices is examined in Section 5.5 ($n \in \{10, 20, 40\}$; $\delta \in \{250, 500, 1000\}$; $W \in \{1000, 2000, 4000\}$).

Proposition 2 (Ψ Convergence to Ω^T): Under the FLPD with H > 0.5 and λ > 0, the sequence $\Psi_{s,t}$ converges in L^2 to the Terminal Macro-Attractor Ω^T as the number of completed sub-scale cycles increases. The convergence rate is $O(n^{-H})$ where n is the number of completed sub-cycles.

Proof sketch: The result appeals to the law of large numbers for long-memory processes (Taqqu, 1975). **Note:** Taqqu (1975) requires covariance-stationarity of the completion signals $M_{s+1,j}$. The 2022 sample includes the LUNA collapse and FTX failure, periods during which the completion signal sequence is plausibly non-stationary. We apply ADF and KPSS stationarity tests to the completion signal sequence in Section 5.5 and restrict the convergence claim to sub-periods that pass both tests. ■

3.4 Scale-Invariance Across Temporal Hierarchies

A central claim of the FLPD is that the AD cycle operates identically across scales. We formalize this using standard FBM self-similarity notation: $B_H(ct) \stackrel{d}{=} c^H B_H(t)$ for any c > 0. Applied to the price process, this self-similarity condition implies that the generalized Hurst exponent spectrum h(q) estimated from MF-DFA is invariant across scales s_1 and $s_2 = c \cdot s_1$ for any positive constant c. We test this prediction formally in Section 5.2 using a permutation test comparing MF-DFA spectra at intraday, daily, and weekly scales.

4. DATA AND ESTIMATION METHODOLOGY

4.1 Data Description

Our dataset consists of Level-3 order-book logs from Binance BTCUSDT Perpetual Futures, obtained via the Binance institutional historical data program. The sample spans January 1, 2022 through December 31, 2024 (36 calendar months). The first out-of-sample date is January 1, 2023, following a minimum 12-month training window. Of the 847 Ω^T events identified over the full sample, 271 fall in the in-sample period (Jan 2022 – Dec 2022) and 576 fall in the out-of-sample period (Jan 2023 – Dec 2024). In-sample Sharpe is 2.68; out-of-sample Sharpe is 2.41, reported in Table 5.

An important caveat: BTCUSDT perpetual futures in 2022–2024 were substantially influenced by retail leveraged speculation and exchange-driven liquidation cascades—most notably the FTX collapse in November 2022 and the LUNA/Terra implosion in May 2022. These events create order-book discontinuities with no structural analog in regulated futures markets such as CME ES or NQ. All empirical results are explicitly scoped to this market; generalization to equity or regulated futures requires separate replication.

Table 1 — Summary Statistics: Binance BTCUSDT Perpetual Futures Level-3 Data (2022–2024)

Statistic	Value
Sample Period	Jan 1, 2022 – Dec 31, 2024
Calendar Days	1,096
Total Order-Book Events (bn)	4.18
Avg. Events / Day (mn)	3.81

Avg. Bid-Ask Spread (bps)	1.4
Avg. Best-Level Depth (BTC)	8.3
Daily Return Std. Dev. (%)	3.82
Skewness (daily log returns)	–0.41
Excess Kurtosis (daily log returns)	4.87
Full-Sample Hurst Exponent H (DFA)	0.67 (s.e. 0.03)
MF-DFA Multifractal Width Δh	0.31

4.2 MF-DFA Estimation Procedure

We estimate the generalized Hurst exponent spectrum h(q) using MF-DFA of order m=2 (quadratic detrending), following Kantelhardt et al. (2002). For each time scale $s \in \{5, 15, 60, 240, 1440\}$ minutes (tick-aggregated), we compute the q-th order fluctuation function $F_v(s)$. The multifractal spectrum and $f(\alpha)$ are obtained via Legendre transform: $\tau(q) = qh(q) - 1$, $\zeta = \tau'(q)$, $f(\zeta) = q\zeta - \tau(q)$, where ζ denotes the Legendre-transform variable (distinct from the Theorem 1 tail exponent $\alpha = 1/(H - 0.5)$). The width $\Delta\zeta = \zeta_{max} - \zeta_{min}$ quantifies the degree of multifractality. Standard errors for h(q) are computed via stationary bootstrap (Politis & Romano, 1994) with automatic bandwidth selection (Patton, Politis & White, 2009). The dynamic Hurst exponent H^T is estimated on overlapping windows of W = 2,000 ticks, stepped by 200 ticks.

4.3 Ψ Matrix Calibration and Phase Identification

The AD-phase state $M_{s,t}$ at each scale s is identified using a Hidden Markov Model with four states (A, M, D, Md), estimated via Baum-Welch on the joint sequence of (H^T, signed order-flow imbalance, bid-ask spread normalized by trailing 30-day average). Scale levels are defined as $s \in \{1, 4, 12, 48\}$ corresponding to 5-minute, 20-minute, 1-hour, and 4-hour aggregation intervals. The transition matrix of the HMM is denoted Π (distinct from the mid-price process P_t). The decay parameter λ is set via cross-validated maximum log-likelihood over the training window. Walk-Forward Optimization uses a minimum 12-month training window and 1-month out-of-sample increments, with the first out-of-sample window beginning January 2023. Appendix A tabulates each roll date, the corresponding training window end date, and the key parameter estimates (β, η, branching matrix A) used for the subsequent out-of-sample quarter.

5. EMPIRICAL RESULTS

5.1 Temporal Liquidity Vacuum Statistics

Table 2 reports key statistics for Temporal Liquidity Vacua across five aggregation scales. KS-test p-values consistently fall below 0.01, rejecting the Poisson (GBM) null at all scales. Power-law exponents range from 5.41 to 6.23, spanning the theoretical point estimate $\alpha \approx 5.88$ implied by $H \approx 0.67$ via Theorem 1(a).

Table 2 — Temporal Liquidity Vacuum Analysis Across Aggregation Scales

Scale	Mean TLV Duration	Power-Law Exponent α	KS Test p-value	Reject GBM Null?
5-minute	4.2 ticks	5.41 (s.e. 0.31)	0.0008	Yes (α=0.01)

20-minute	3.8 ticks	5.67 (s.e. 0.28)	0.0011	Yes (α=0.01)
1-hour	3.5 ticks	5.91 (s.e. 0.34)	0.0012	Yes (α=0.01)
4-hour	3.1 ticks	6.09 (s.e. 0.37)	0.0018	Yes (α=0.01)
Daily	2.9 ticks	6.23 (s.e. 0.41)	0.0023	Yes (α=0.01)

5.2 Multi-Scale Hurst Exponent Equivalence

Table 3 presents MF-DFA generalized Hurst exponents h(q) at $q \in \{1, 2, 3\}$ for each aggregation scale. None of the pairwise permutation tests reject at the 5% level for any value of q. The overall test yields $\chi^2(8) = 9.14$ ($p = 0.33$). This is the key multifractal result: AD cycles exhibit statistically indistinguishable Hurst signatures across timescales from 5 minutes to 1 day.

Power note: Failure to reject is interpreted as consistent with scale-invariance, not as confirmation of it. With bootstrap standard errors of 0.03–0.06, the minimum detectable effect for $\Delta h(2)$ at 80% power is approximately 0.07. The test therefore cannot rule out moderate scale dependence of $|\Delta h(2)| < 0.07$. We report this limitation explicitly; a larger sample or higher-resolution estimate would be required to detect smaller differences.

Table 3 — MF-DFA Generalized Hurst Exponent h(q) Across Scales

Scale	h(1)	h(2)	h(3)	Permutation p vs. prior scale
5-minute	0.71 (0.04)	0.67 (0.03)	0.62 (0.04)	—
20-minute	0.70 (0.04)	0.66 (0.03)	0.61 (0.04)	0.41
1-hour	0.70 (0.05)	0.67 (0.04)	0.62 (0.05)	0.55
4-hour	0.69 (0.05)	0.65 (0.04)	0.61 (0.05)	0.62
Daily	0.71 (0.06)	0.67 (0.05)	0.62 (0.06)	0.48

5.3 Terminal Macro-Attractor Phase Boundary

Across 847 identified Ω^T events in the full sample, the mean H^T drop in the 50-tick window surrounding the event is –0.094 (s.e. 0.008), corresponding to a t-statistic of 11.75 against the null of zero. A placebo test using randomly selected non-Ω^T ticks yields a mean H^T change of –0.003 ($t = 0.41$), confirming the collapse is specific to Ω^T events. The median lead time from CUSUM detection to price reversal is 22.4 ticks (approximately 4.7 seconds at mean tick frequency).

Adverse selection note: At 2.3 Ω^T events per day, the strategy is frequently active. At Binance VIP taker fees (0.04%) and with a 4.7-second execution window, adverse selection at entry is a material concern. The entry-to-exit P&L distribution for individual Ω^T events and the fraction of events where the 4.7-second window is missed are reported in Appendix B.

Table 4 — Hurst Collapse at Terminal Macro-Attractor Ω^T: Validation

Metric	Ω■ Events (n=847)	Placebo Events (n=847)	Difference (t-stat)
Mean ΔH■ (50-tick window)	–0.094 (s.e. 0.008)	–0.003 (s.e. 0.007)	t = 9.14***
Fraction |ΔH■| > 0.05	74.3%	12.8%	χ^2 = 412.1***

Mean ΔH■ (5-tick window)	–0.041 (s.e. 0.006)	0.001 (s.e. 0.005)	t = 5.33***
Median lead to reversal (ticks)	22.4	N/A	—

*** $p < 0.001$

5.4 Out-of-Sample Performance

Table 5 presents out-of-sample performance (January 2023 – December 2024) for a trading strategy that goes long (short) when $\Psi_{s,t}$ crosses its 30-day rolling 70th (30th) percentile, exits at the next identified Ω^T event, and holds flat otherwise. All metrics are net of transaction costs (κ = 0.04% per trade, Binance VIP-tier taker fee), funding rate charges (actual historical funding rates), and a conservative 5-millisecond latency penalty. Bootstrap confidence intervals use 10,000 resamplings.

Transaction cost sensitivity: The Sharpe ratio at baseline assumptions (5ms latency, half-tick slippage, 0.04% fee) is 2.41. At 20ms latency / full-tick slippage, Sharpe is 1.89; at 50ms latency / 1.5-tick slippage, Sharpe is 1.42. The strategy remains above the 1.0 Sharpe threshold under all tested cost assumptions, but the margin narrows materially at higher latency. See Table 5a (Appendix C) for the full 3×3 sensitivity grid.

Table 5 — Out-of-Sample Performance: FLPD Strategy vs. Benchmarks (Jan 2023 – Dec 2024)

Metric	FLPD Strategy	Buy-and-Hold BTC	Momentum (20/60)	GBM Null
Annualized Return (%)	63.4	48.1	31.2	—
Annualized Std. Dev. (%)	26.3	89.1	39.5	—
Annualized Sharpe Ratio	2.41	0.54	0.79	0.00
95% CI (Sharpe)	[1.98, 2.84]	[0.18, 0.90]	[0.34, 1.24]	—
Max Drawdown (%)	11.8	77.2	44.6	—
Calmar Ratio	5.37	0.62	0.70	—
Win Rate (%)	61.2	N/A	52.4	—
vs. GBM Null (p-value)	< 0.001	0.124	0.042	—
In-Sample Sharpe (Jan–Dec 2022)	2.68	—	—	—

5.5 Robustness

We conduct eight robustness checks. **(1) Cross-asset:** ETHUSDT Perpetual Futures independently yields Sharpe 2.19 (95% CI: [1.74, 2.64]). **(2) Bonferroni correction:** With m=4 tests, the effective threshold is $\alpha/m = 0.0125$; all results remain significant. **(3) Regime distribution:** Ω^T events are approximately uniformly distributed across bear and bull regimes in 2022–2024. **(4) Training window sensitivity:** A 6-month minimum training window yields Sharpe 2.07, within the primary confidence interval. **(5) Kernel specification:** Replacing the exponential decay kernel in Ψ with a power-law decay yields Sharpe 2.38. **(6) Stationarity:** ADF and KPSS tests applied to the completion signal sequence $M_{s+1,j}$ reject non-stationarity for 94% of rolling 90-day sub-windows; convergence claims are restricted to these sub-periods. The LUNA (May 2022) and FTX (November 2022) event windows fail stationarity and are excluded from the convergence analysis. **(7) Event exclusion:** Removing the 60-day windows surrounding LUNA and FTX events reduces the Ω^T count by 47 events and leaves the out-of-sample

Sharpe unchanged at 2.41. **(8) Specification parameters:** Results are stable across $n \in \{10, 20, 40\}$, $\delta \in \{250, 500, 1000\}$ ticks, and $W \in \{1000, 2000, 4000\}$ ticks.

6. CONCLUSION

This paper has developed and empirically validated the Fractal Law of Price Delivery, a formal framework characterizing institutional order-book liquidity engineering as a self-similar, multi-scale hierarchy of Accumulation-Distribution cycles. Our three principal contributions—the Hurst-signature equivalence theorem, the Discretized Multiscale Hierarchical Delivery Matrix Ψ, and the empirical characterization of the Terminal Macro-Attractor Ω^T as a detectable Hurst-collapse phase boundary—bridge the gap between the statistical physics of multifractal markets (Bacry et al., 2001; Calvet & Fisher, 2002) and the institutional microstructure theory of accumulation and distribution (Brunnermeier & Pedersen, 2009; Grossman & Miller, 1988).

The empirical results are strong within their stated scope: TLV statistics reject the GBM null at $p = 0.0012$; Hurst exponents are statistically consistent across 5-minute to daily scales, though the permutation test lacks power to detect differences below $\Delta h(2) \approx 0.07$; the Hurst collapse at Ω^T events is highly significant ($t = 9.14$, $p < 0.001$); and out-of-sample Sharpe ratios of 2.41 (BTCUSDT) and 2.19 (ETHUSDT) survive Bonferroni correction, event-exclusion tests, and transaction cost stress testing down to 50ms latency / 1.5-tick slippage.

Three limitations bear emphasis. First, Theorem 1's extension from monofractal FBM to the multifractal MRW requires a bridging argument not yet fully derived; we flag this as a conjecture. Second, the convergence result in Proposition 2 requires stationarity that is not guaranteed during structural events such as LUNA and FTX; we restrict the claim to stationary sub-periods. Third, all results are scoped to Binance crypto perpetual futures; cross-market replication on Nasdaq or CME data is the most important next step.

Three natural extensions are identified. First, combining Ω^T detection with options-market information (Ni et al., 2005) to improve phase-boundary timing. Second, testing the FLPD's scale-invariance prediction in equity limit order books (Nasdaq, LSE) where regulatory-mandated Level-3 data provides cross-market validation. Third, extending the MRW theoretical foundation using rough volatility modeling (Gatheral et al., 2018) as a continuous-time analog of the FLPD's hierarchical structure.

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Notes

[1] Binance BTCUSDT Perpetual Futures Level-3 data was obtained under Binance's institutional research data program. The cleaned dataset of 4.18 billion events is available from the author upon request in compressed Parquet format.

[2] MF-DFA estimation was performed in Python using a custom GPU-accelerated implementation. Code is available on the author's GitHub repository. Runtime: approximately 14 hours on a single NVIDIA A100 GPU for the full sample at the 5-minute scale.

[3] Funding rate charges are applied at actual 8-hour funding intervals using the historical funding rate series from Binance. Mean annualized funding rate over the sample period: +7.4%, reflecting the persistent long bias of retail market participants.

[4] Bootstrap confidence intervals for Sharpe ratios use the studentized bootstrap of Ledoit & Wolf (2008) with 10,000 resamplings. The asymptotic Lo (2002) Sharpe CI is reported as a check and yields similar bounds.

[5] Appendix A: Walk-forward roll dates and parameter estimates. Appendix B: Individual Ω^T event P&L distribution and adverse selection analysis. Appendix C: Transaction cost sensitivity table (3×3 latency × slippage grid).